VoiceOps 2026 Capstone Report

ZentroMart customer-support agent | Evaluation and submission status | Reference date: 20 October 2026

What was built

A four-step support agent: Step 1 prompting with Gemini; Step 2 document-grounded retrieval with source IDs; Step 3 conversation memory, verified order lookup and ticket creation; Step 4 Gnani-only speech-to-text and text-to-speech. The local website exposes Steps 1-3 and a Step 3 microphone flow, transcript, spoken reply and retrieved sources while reusing the same Python agent. Optional OpenAI-compatible embeddings and the in-memory vector index remain unchanged. Synthetic dataset only.

Evaluation results

The Step 1/2 scores below are from the saved historical run that used question text. The evaluator now defaults to the brief's asr_text field; these scores were not refreshed during this audit to conserve Gemini quota.

Stage	Cases	Required facts	Evidence	Tool checks
Step 1 prompt-only	52	16.3% (8/49)	n/a	52/52
Step 2 grounded RAG	52	85.7% (42/49)	100% (47/47)	52/52
Step 3 single questions	52	81.6% (40/49)	100% (47/47)	52/52
Step 3 ASR-like text	52	71.4% (35/49)	93.6% (44/47)	52/52
Step 3 conversations after fixes	96	76.6% (59/77)	100% (96/96)	100% (96/96)
Step 4 live synthetic audio	3 turns + silence	Manual review	Sources saved	Ticket confirmed

Scores use automated string and source checks; review paraphrases, equivalent dates and numbers, and Hindi/Hinglish quality. The conversation benchmark improved from 76/96 tool checks and 45/96 evidence checks to 96/96 on each after fixes. Required-fact screening remains 59/77.

What went wrong and what changed

Misses came from lost or guessed order IDs, missing verification and late-refund policy sections, and a skipped human ticket. The agent now binds lookups to a caller-supplied or pending ID, retrieves failed-verification guidance, escalates overdue verified refunds using dataset date rules, and confirms tickets only after created=true. A regression fix also treats a bare order ID as a reply to a plain request for an order ID, then uses the pending order for the caller pincode or phone suffix.

Step 1 to Step 2 required-fact screening rose from 16.3% to 85.7%; Step 2 matched 47/47 scored evidence checks. These automated scores are screening signals, not a replacement for human review.

Voice and reproducibility

A three-turn Gnani STT/Gemini/Gnani TTS batch and a separate silent-audio recovery case passed in an earlier run; neither was repeated in this audit. The website text, voice and HTTP routes pass mocked offline tests; real browser microphone permission and live website audio remain untested. Offline tests: 35 passed. Dataset validation passed all nine CSVs, 21 knowledge documents, joins, order math, practice evidence and verification gates.

Batch command: python batch_run.py <audio-folder> --output sample_outputs | Website: python web_app.py (http://127.0.0.1:8000) | Text: python run.py --step 3. Repository URL: not published yet. The ZIP excludes .env, API keys, virtual environments and dataset copies. Gemini request spacing is set to 4.1 seconds for a 15-requests-per-minute free tier.